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Advancing Cloud Gaming: Addressing Latency Challenges through Research and Innovative Solutions

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***Abstract*— Cloud gaming is changing the gaming industry by offering high-quality gameplay without needing powerful local hardware. However, common problems like high lag, network congestion, and inconsistent Quality of Experience (QoE) limit its broader use. This document addresses these challenges through an in-depth analysis of a large-scale latency dataset gathered from 189 players interacting with 9 servers across North America, South America, Europe, and East Asia. Utilizing several models, such as Random Forest, Gradient Boosting, and Linear Regression, we predict latency patterns and identify the elements influencing performance declines. Our results reveal significant regional variations in server performance and show that cloud gaming performance may be significantly enhanced by hybrid configurations that include the best features of both platforms, such as adaptive Quality of Experience (QoE) models and local computation through edge nodes. Three key contributions are provided by this research: (1) a data-driven model for predicting latency in global cloud gaming networks; (2) a detailed analysis of leading platforms to ascertain their advantages and drawbacks; and (3) the proposal of a novel hybrid framework based on data-driven resource allocation techniques. These contributions lay the groundwork for improving user experience across a range of network configurations and optimizing cloud gaming systems.**

Keywords—Cloud Gaming, Latency, Bandwidth, Video Codec, Edge Game, Gaming Anywhere, Virtualization, QoE, Distributed Systems

I. INTRODUCTION

An invention called cloud gaming, which is a new business model that has captured the imaginations of both global firms in the industry and individual gamers in recent years, has had a massively favorable impact on the growing video game industry. Cloud computing resources can be utilized across a range of applications, with resource-intensive computer games emerging as a standout use case for cloud computing [1]. In the context of mobile cloud gaming, the user interface design becomes crucial in shaping the user experience, particularly when the streamed games were initially developed for platforms other than mobile.

Thus, the fast advancement of cloud settings and new legal aspects of the environment at the skinny customer period can be discussed with reference to the following principal innovations related to cloud gaming platforms. Currently, many game companies have developed different game platforms such as OnLive, StreamMyGame, and Gaikai that have revolutionized the cloud gaming through practicing and experimenting on different trends and strategies that meet the different demands and needs of gamers in the world. Among these challenges has been formerly addressed by these platforms such as software installation overhead, compatibility issues with hardware, and the repetitiveness in hardware upgrades through the scalabilities of the cloud-based assets hence providing an excellent movie review to the intended audience [2].

The shape of this paper is as follows: We start our work by providing literature review that would further stand the study in the framework of cloud gaming research. We proceed to characterize a robust methodology for quantifying latency in closed structures, given the methodological challenges networked by means of proprietary structures. We then go further to analyze and compare various top cloud gaming platforms like OnLive and StreamMyGame which are

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explained here in detail about the streaming latency, total performance, and consumer pleasure of the systems. At long last, we provide suggestion and a possible perspective for another research: The continued investigations are crucial for the development of cloud gaming in the future.

II. Background

Cloud gaming is becoming increasingly popular. For instance, CloudUnion faces a significant number of subscribers exceeding its current infrastructure capacity. To address issues like prolonged waiting times, an admission control algorithm was introduced [4]. This approach highlights the growing demand for on-demand or cloud gaming services.or game streaming is a type of technology by which without having a gaming console one can play games that are delivered online from servers that can be away from such users at very large distances. Web-based means that there are no applications installed in clients, computers, or other devices such as console or PC, instead, the powerful servers describing data centers handle all the game processes. Controlling flows are then sent to the game to produce the display of the user in real time such as the images to be displayed, the sound to be played, the signal for keyboard click or the movement of the game pad are then sent back to the server for interpretation or for a computation to be made. This configuration makes it possible for games to be played on virtually all platforms including pc even with low config, tablets, smart phones, and smart TVs since most of the computation is done on the server. This technology is exemplified by current practices such as NVIDIA GeForce NOW, Google Stadia, Microsoft Xbox Cloud Gaming, and PlayStation Now. There are a number of benefits when it comes to on-demand gaming: firstly, the costs of overall gaming is slightly reduced, and secondly, the ownership of consoles is not necessary in order to play the games on demand without needing to download or install them. It also has its disadvantages: high server response time, bandwidth I/O, and significant data usage; these factors impact the game performance and might be inconsiderate towards the players with poor connectivity or a restricted data plan. The video game industry has played a significant role in the software and entertainment sectors. For example, the global video game market is predicted to increase from 66 billion US dollars in 2010 to 81 billion US dollars in 2016 **[6]**. Another market research study **[7]** further breaks down the market growth into three categories: boxed-games, online-sold games, and cloud games.

III. Literature Review For Latency Measurement

A) Transparent- gaming

A comprehensive description of the processes explored in the study and a detailed discussion of the different strategies applied towards reducing latency challenges in cloud gaming solution is outlined, particular through the use of T-Gaming platform. With this performant separation of the latency factors, the work endeavors to expound more on processing delay (PD), playout delay (OD) and the network delay (ND) which are Among the most delectable factors that causes latency to user. Out of the flow of implementation methods inclusive of GPU, CPU-software, HWA and method of benchmarking, the undertaking finds out the best possible way of avoiding the delay in processing and frame encoding. Further, the research aims at investigating advance streaming strategies with the incorporation of the enhanced DRL algorithms to adaptively fine-tune the bitrate based on real-time network conditions.

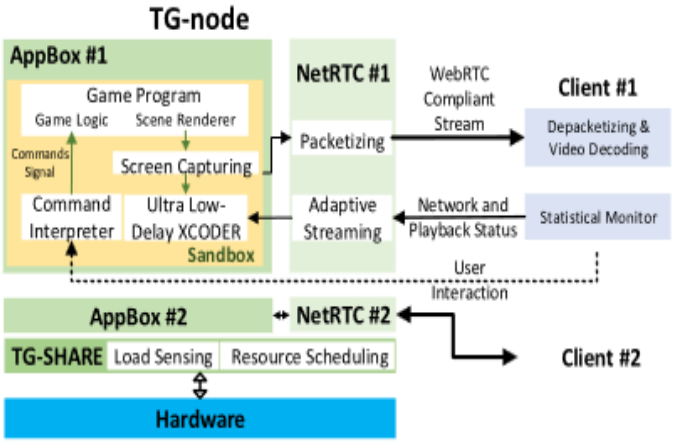


Fig. 1. Game Cloud Design with Virtualized CPU/GPU Servers and Initial Performance Results (Zhao, Hwang, & Villeta, 2012).

This technique developed in the course of this study known as ARS outperforms the conventional approaches of congestion control; because it has the ability of dynamically dominating bitrate based on this real-time network conditions in order to enhance the Quality of Experience (QoE) for these games. In fact, considering the experiment of the study, ARS is trained with the help of various simulated network traces of real-world datasets and consequently it realizes that ARS offered superior average QoE and stability in different types of networks.

B) Edge Game

Edge Game improves business visions and perceptions fundamentally of cloud gaming by deploying computation-intensive GPU rendering near the edge nodes, curbing latency and usage of bandwidth. It means that the game processes and graphics are managed near the user, far from latency that negatively affects the games hosted in the cloud. It takes advantage of machine learning techniques to dynamic video encoding and control of network throughput for the purpose of achieving optimum video quality on the basis of the existing physical network standards.

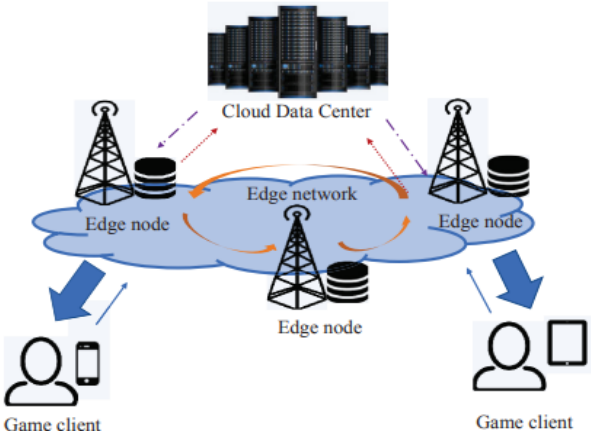


Fig. 2. Framework of EdgeGame (Xu Zhang, Hao Chen, 2019).

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The technological base of the system is virtual nodes (vNodes) with relatively high levels of game processing on edge devices, while user accounts, system diagnostics, and the configuration of game sessions are held in a cloud data center.

C) Fog Computing

Due to the popularity of massively multiplayer online games (MMOGs) and the growth of mobile gaming, certain drawbacks of traditional cloud gaming models—such as latency and coverage—are becoming more apparent. **[10]** Due to the distance between the players and the servers, traditional cloud gaming uses large server farms and fast internet connections, which can lead to latency issues. This latency is detrimental to the user's quality of experience **(QoE)**, especially when combined with high bandwidth consumption. **[11] [12]** Prior studies have indicated that increasing the number of datacenters may be one way to address these issues, but doing so will come at a cost.

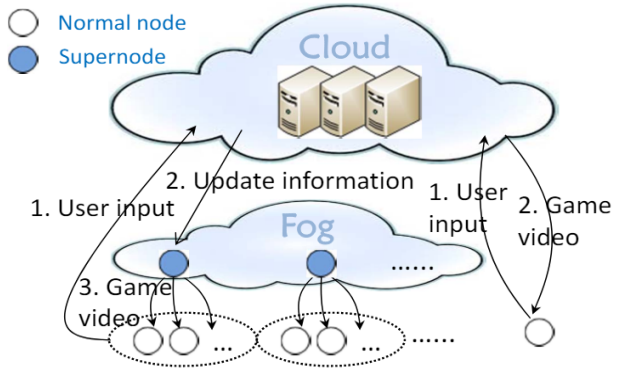


Fig. 3. Architecture of FOG Computing (N. Tolia, D. Andersen, 2019)

In order to tackle these issues, the concept of CloudFog has been proposed. Another "fog" layer with supernodes located closer to the consumers is suggested by CloudFog. While the cloud performs the calculations for the game updates, these supernodes manage the rendering of the game videos and broadcast them to the players**.** By reducing the quantity of data sent from the cloud to the end users, CloudFog reduces response time and bandwidth usage **[17]**.

D) Network Latency Formulation

Network latency is the sum of all possible delays a packet can face during data transmission. We generally express network latency as round trip time (RTT) and measure in milliseconds (ms). Network delay includes processing, queuing, transmission, and propagation delays. Let's look at the formula to calculate network latency:

$$NL = DP + DQ + DT + DPR \quad (1)$$

Transmission delay is the time to push all the available data in a transmission medium or wire. We can calculate transmission delay (in a second) by dividing the number of bits by the transmission rate:

$$DT = \frac{NB}{TR} \quad (2)$$

Another delay a packet might face during data transmission is propagation delay. It's the time taken for a packet to cross the transmission medium. It depends on the distance (D) and speed of the packet (S). Hence, propagation delay:

$$DPR = \frac{D}{S} \quad (3)$$

We can calculate network throughput (TP) using TCP receive window (W) and network round trip time (RTT) which is related to latency:

$$TP \leq \frac{W}{RTT} \quad (4)$$

IV. Methodology

The approach to research used in this study includes formulation of research questions that follow a systematic approach that deals with latency prediction in network communications. It covers steps such as data gathering, data cleaning, selecting algorithms, building models, assessment and interpretation. Each of the steps is highly worked out in order to achieve the high degree of measurability to express latency, which is one of the key area defined for network performance assessment.

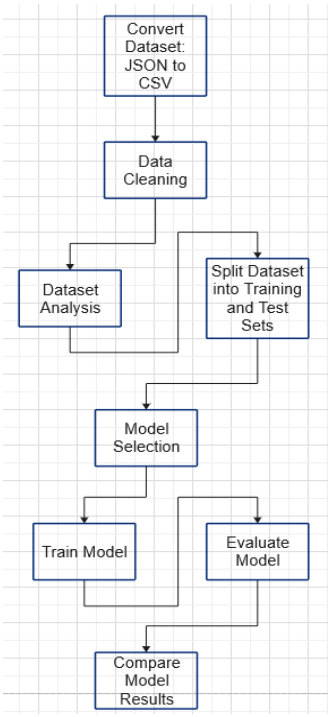


Fig. 4. Machine Learning Flowchart

A) Dataset Collection

The dataset utilized in this study was sourced from IEEE Dataport, comprising network delay measurements between 189 players and 9 servers distributed across 4 regions: Producers from North America, South America, Europe and East Asian countries [19].

B) Dataset Conversion

The data was provided in JSON database format and with the help of Python language it was converted to CSV format. The JSON records contained network delay measurements from each player to the servers, with additional data such as:

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The JSON records contained network delay measurements from each player to the servers, with additional data such as:

Algorithm 1: Parse JSON Data and Create CSV

```
Initialize an empty list records;
foreach entry in data:
    Extract key values (user_id, latitude, longitude, asn_org, timestamp);
    foreach server_stats in entry['data']['stats'] if valid;
        Create and append record with extracted values and latency stats (min, max, avr);
    end
end
Convert records to a Dataframe df;
Save df
```

C) Dataset Cleaning

It was important to clean and transform the data in order to acquire a sound and accurate dataset before engaging in the modeling process. Data cleaning is a vital step in data preparation that was performed on the dataset by undertaking several vital processes to fix numerous issues such as inconsistent values, missing values, and outliers.

Algorithm 2: Preprocess Data

```
Drop the Unnamed: 0 column from data;
Convert latitude column to numeric,handling errors;
Remove any rows with NaN values in data
foreach column in ['user_id ', 'asn_org ', 'server ']:
    Apply label encoding and store encoders;
end
```

D) Dataset Analysis

Our Analysis highlight serious shortcomings in the existing cloud gaming setup, especially with regard to network latency. Based on the player's location, the ISP they use, and the server they connect to, measurements show that latency varies significantly. Remarkably, connection testing was consistently conducted only in the United Kingdom (uk), suggesting a possible bottleneck in evaluating the entire range of infrastructure performance.

E) Distributions by Server

This type of map helps visualize how latency varies geographically across user locations. Areas with darker or more intense colors represent higher average latency, while lighter colors indicate lower latency. By looking at the color distribution, you can identify regions or countries with generally higher or lower latency. This might be helpful in understanding how geographical factors influence user experience.

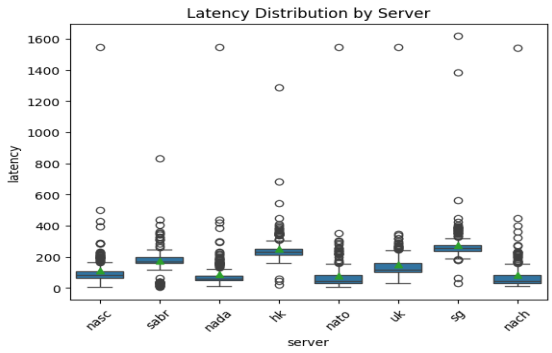


Fig. 5. Latency Distribution based on Server

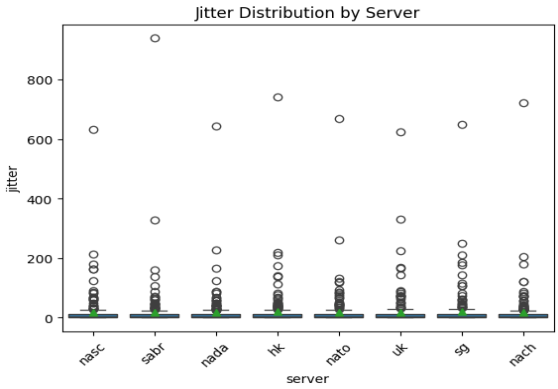


Fig. 6 Jitter Distribution based on Server

F) Dataset Spliting

We split our dataset for training and testing almost 80% of data is for training and 20% for testing. It is a basic rule in the machine learning process.

Split Dataset for Training and Testing

- Separate the target variable (latency) from the input features in the dataset.
- Split the dataset into two parts:

Training Set: 80% of the data for model training.

Testing Set: 20% of the data for evaluating the model.

G) Models Selections

Three distinct regression models were considered for this study: Linear Regression, Random Forest and Gradient Boosting among others. Each model was evaluated not only by its intrinsic characteristics and the ability of the model to fit regression problems but also by its ability to fit the data characteristics of latency distribution.

Algorithm 2: Data Splitting, Training and Evaluation

```
Define features X as data without latency, and target y as data['latency'];
Split X and y into training and testing sets (X_train, X_test, y_train, y_test) with an 80-20 split;
Initialize dictionaries results and predictions;
foreach model in models:
    Train the model on X_train and y_train;
    Predict y_pred using X_test;
    Evaluate performance using MAE (mean square error) and R^2 score;
    Store results (MAE and R^2) and predictions;
```


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End

H) Models Training and Evaluation

During model training and evaluation, we use the split-input on the dataset, where 80% of the data was utilized for training purposes and 20% of the data as the testing purpose. The linear regression model was proved to be the most accurate model among all of the models as it got the lowest Mean Absolute Error (MAE). Behind the decision trees is the Random Forest model while Gradient Boosting is slightly behind the Random Forest.

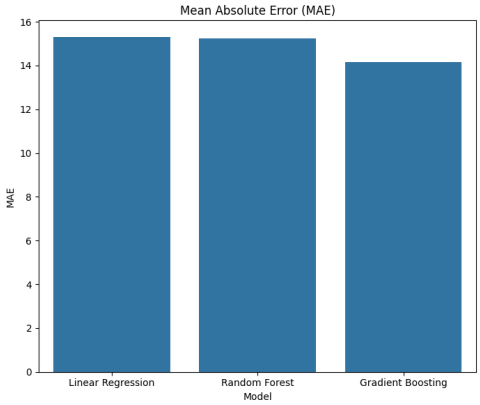
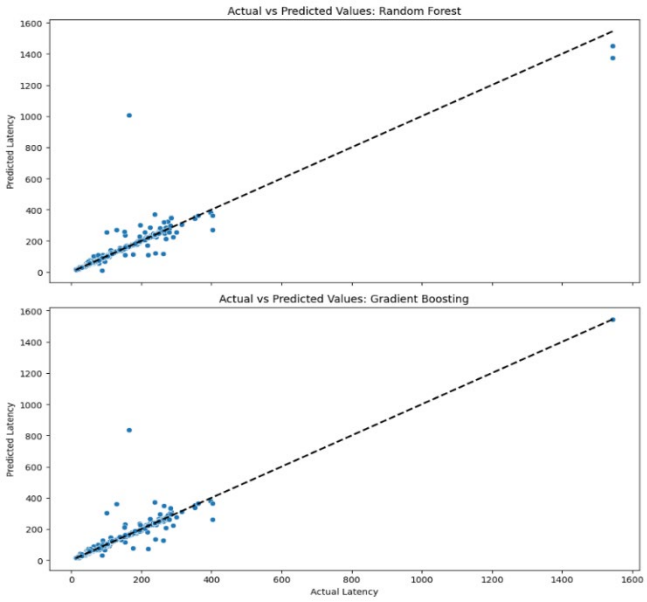


Fig. 7. Models Training and Evaluation Code Snapshot

I) Actual vs Predicted Values Comparison

To show the relation between two variables, we use scatter plots. On these graphs, actual latency values are represented by x-axis; predicted latency values are represented by y-axis. Each data point represents a prediction.In all three of the graphs above, most of the points appear to cluster around a straight line, albeit with some scattering.



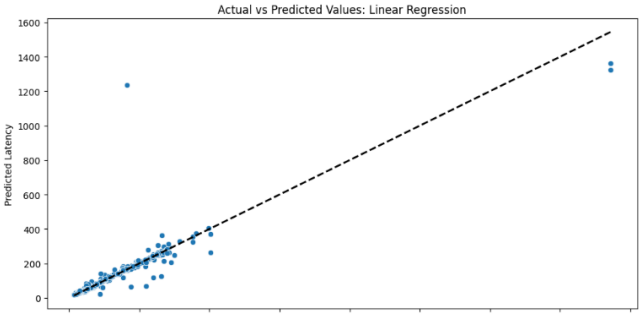


Fig. 8. Actual vs Predicted Values between 3 models

This shows that for many of the data points, the models can be quite accurate in their predictions. However, it is also evident that several points lie far away from this line meaning that these models were not correct in predicting them.

V. Infrastructure Limitations

The amount of servers currently in use is not enough to offer a worldwide user base low-latency gaming experiences. Real-time gameplay may be hampered by the high latencies and noticeable jitter seen, which would make for a frustrating user experience. These problems are especially noticeable in areas with lower network infrastructure and fewer server deployments.

VI. Potential Solutions

In order to tackle these obstacles, various approaches could be utilized, such as:

- Increasing Server Density: By putting more servers in different places, you can physically close the distance between players and servers, which will cut down on latency. This strategy would entail adding more servers to underserved areas, like parts of South America and currently of Asia. EDGE GAME can be the best fit for this purpose.
- Optimizing Server Location: By carefully placing servers in relation to user density and network performance information, latency can be decreased and coverage can be improved. Using advanced algorithms to analyze player distribution and network conditions can help identify optimal server locations.
- Enhancing GPU Virtualization: More concurrent game instances can be supported by enhanced GPU virtualization technologies, which will increase the performance and scalability of cloud gaming services. To guarantee exclusive access to resources, each game instance is given its own virtual GPU and GPU memory.
- Developing Detailed (QoE) Models: The gaming experience can be enhanced by developing sophisticated Quality of Experience (QoE) models that dynamically adjust system parameters based on network conditions and player device capabilities. These models can use learning algorithms to adjust video coding bitrate and frame sampling rate in real-time.

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VII. Conclusion

Cloud gaming is a rapidly growing subject that has a lot of potential but also has major drawbacks such as high latency and excessive resource usage. Our study aimed to address these problems by predicting latency patterns and identifying performance-affecting factors using machine learning models (Linear Regression, Random Forest, and Gradient Boosting). By using a data-driven approach, we were able to identify the precise circumstances that lead to latency spikes, which made it possible to improve network management techniques. All things considered, incorporating content-specific codecs, sophisticated GPU virtualization, and optimized hybrid frameworks such as CloudFog greatly improves the Quality of Experience (QoE) for users. As network technology and hardware solutions continue to advance, like NVIDIA's GeForce Grid, cloud gaming is expected to surpass its existing constraints and offer more immersive experiences.

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